

Fig. 2: Sub-grid set-up for a mobile phone simulation in the presence of head and hand

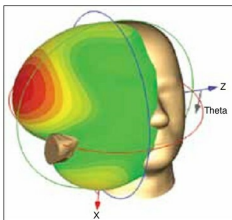


Fig. 4: Far-field pattern at 1800 MHz

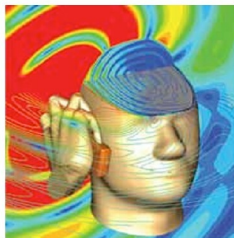


Fig. 3: E-field distribution at the GSM band (1800 MHz)

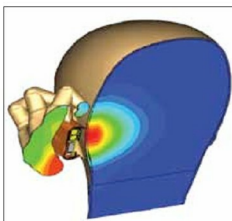


Fig. 5: SAR distribution averaged over 1 gm following the IEEE C95.3 standard

the wavelength inside the head is shortened compared to the vacuum, due to high dielectric values. The radiated field is basically guided around the head.

The far-field plot (Fig. 4) shows that the main radiation direction is located between the hand and the head.

For certification of the mobile phone, SAR is expected to be compliant to the SAR standard IEEE C95.3. Currently, the SAR value needs to be measured, but in the design stage it is very useful to check this value through simulation. In the near future, pure simulation may be sufficient for certification.

The 'SAR 10 gm' is the maximum SAR value averaged on 10 gm, which is obtained by averaging the SAR around each point in the volume and adding the nearest points till an average mass of 10 gm is reached with a resulting volume having the shape of a portion of sphere. The contiguous SAR 1 gm is estimated by averaging the local maximum SAR, adding the highest SAR volume in a given tissue till a mass of 1 gm is reached. SAR (point) is the local value of SAR at every point inside the head model.

Fig. 5 shows distribution of the SAR value averaged over 1gm tissue mass. It is clear that the radiated field is basically guided around the head and varies inversely with distance. This leads to an automatic inference that SAR and hence the health effects within very close proximity of head would be harmful. Thus it is clear that a person using a mobile phone 30-40 cm away from his body will have a much lower exposure to radio-frequency fields than someone holding the handset against his head. So it is recommended to keep mobile phones away from your body as much as possible. ●

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Simulation of the mobile phone, with head and hand models, requires tens of millions of mesh cells and a simulation run-time of five to six hours on a PC (Intel Core i3 CPU, 64-bit operating system, 2.13GHz, 4GB RAM). The number of mesh cells can be re-

duced using the sub-gridding scheme. When applied, a very fine mesh is created inside the phone, a coarser one in the head and a very basic one in the vacuum (Fig. 2).

Due to high dielectric permittivity of the order of 40 in the brain tissue, the mesh needs to be much finer in the head region than the surrounding vacuum. In classical time-domain solvers, these fine mesh lines need to be extended to the boundary of the computational domain. However, the sub-gridding scheme in CST MWS allows different levels of mesh to be used for different solids in the model. Its use reduces the number of mesh cells required and also the simulation run-time.

Such a simulation can give an important insight into how the SAM phantom or the homogeneous body models affect the performance of the mobile phone. The resulting electric field distribution at 1800 MHz is shown in Fig. 3. It clearly shows how